3.

Let There Be Light

After the big bang, the main agenda of the cosmos was expansion, ever diluting the concentration of energy that filled space. With each passing moment the universe got a little bit bigger, a little bit cooler, and a little bit dimmer. Meanwhile, matter and energy co-inhabited a kind of opaque soup, in which free-range electrons continually scattered photons every which way.

For 380,000 years, things carried on that way.

In this early epoch, photons didn't travel far before encountering an electron. Back then, if your mission had been to see across the universe, you couldn't. Any photon you detected had careened off an electron right in front of your nose, nano- and picoseconds earlier.[†] Since that's the largest distance that information can travel before reaching your eyes, the entire universe was simply a glowing opaque fog in every direction you looked. The Sun and all other stars behave this way, too.

As the temperature drops, particles move more and more slowly. And so right about then, when the temperature of the universe first dipped below a red-hot 3,000 degrees Kelvin, electrons slowed down just enough to be captured by passing protons, thus bringing full-fledged atoms into the world. This allowed previously harassed photons to be set free and travel on uninterrupted paths across the universe.

This "cosmic background" is the incarnation of the leftover light from a dazzling, sizzling early universe, and can be assigned a temperature, based on what part of the spectrum the dominant photons represent. As the cosmos continued to cool, the photons that had been born in the visible part of the spectrum lost energy to the expanding universe and eventually slid down the spectrum, morphing into infrared